

Model Experimentation & Ablation Report

TGSR Project 1B

Empirical status: engineering smoke test on synthetic data. These numbers are NOT historical NSE results.

1. Executed model comparison

The reference implementation executes GCN, GraphSAGE, GAT, GATv2 and TGAT. The table below is the actual output of the smoke-test run.

model	IC	RankIC	NDCG@3	Top3HitRate
GCN	0.0186	0.0326	0.3736	0.3106
GraphSAGE	-0.0065	-0.0272	0.2999	0.2197
GAT	0.05	0.0505	0.3043	0.2197
GATv2	0.0223	0.0083	0.3272	0.25
TGAT	-0.0053	-0.0177	0.2713	0.1894

2. Executed ablation smoke test

These ablations are executed on the same synthetic data and split. The production empirical programme must additionally run the feature-family, edge-type, heterogeneous/signed and Optuna experiments specified by the brief.

Configuration	IC	RankIC	NDCG@3	Top3HitRate	Delta_IC
Full TGAT	-0.0053	-0.0177	0.2713	0.1894	0.0
No temporal (GATv2)	0.0223	0.0083	0.3272	0.25	0.0276
Single head	-0.0424	-0.0494	0.2556	0.1742	-0.0371

3. What this validates

The smoke test validates code execution, metric generation and reproducibility. It does not validate economic performance, historical sector behaviour, or superiority of any architecture in Indian markets.

4. Production experiment plan

Run nested expanding-window CV with Optuna; log every trial; compare IC, Rank IC, NDCG and top-k hit rate; then run calibration, explanation stability, robustness, transaction-cost backtest and audit-lineage checks.

5. Stop/keep rule

Retain complexity only where removal produces a meaningful and stable out-of-sample deterioration under identical folds. Unstable explanations are flagged rather than presented as facts.

Appendix 1: Reproducibility checklist

Record data snapshot hash, code commit, configuration, random seed, fold IDs, training log, attention artifact, uncertainty calibration artifact and post-trade attribution.

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